

Roll no: 26k-0006

PART B:

SYSTEM DESIGN SCENARIO

Ques no 1 (Hotel Booking system)

Algorithm:

- 1) Start
- 2) Initialize Total-Hotel-Revenue = 0
- 3) Set number of guests $N = 4$.
- 4) For Guest = 1 to N
 - a) Input Season (Peak/Off Peak), Room-Type (standard/Deluxe / Suite), and Nights (Number of nights)
 - b) Check season
 - IF Season is "Peak".
 - IF Room-Type is "Standard", set Rate = 5000.
 - Else IF Room-Type is "Deluxe", set Rate = 8000.
 - Else IF Room-Type is "Suite", set Rate = 12000
 - Else IF season is "Off Peak".
 - IF Room-Type is "standard", set Rate = 3000.
 - Else IF Room-Type is "Deluxe", set Rate = 5000.

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- Else IF Room-Type is "Suite", set Rate=8000.
- 5) Calculate Initial Bill: $\text{Initial Price} = \text{Rate} * \text{Nights}$.
- 6) Check for Discount
 IF Nights > 7
 Calculate Discount = $\text{Initial Price} * 0.15$,
 Else
 Set Discount = 0.
- 7) Calculate Final Price:
 $\text{Final Price} = \text{Initial Price} - \text{Discount}$
- 8) Display Final-Price.
- 9) Accumulate to total revenue:
 $\text{Total-Hotel-Revenue} = \text{Total-Hotel Revenue} + \text{Final-Price}$
- END LOOP
- 10) Display Total-Hotel-Revenue
- 11) Stop

Program Analysis Chart (PAC)

Given Data	Processing Required	Output Requirement
• N=4 (Number of guests)	• find Rate based on Season and Room-Type.	• Final-Price for each guest
Season (Peak/off Peak)	• Initial Price = Rate * Nights	
Room-Type (standard, deluxe, suite)	• If Nights > 7 , Discount = Initial Price * 0.15 else Discount = 0.	• Total-Hotel-Revenue
Nights (No. of nights stayed)	• Final-Price = Initial Price - Discount	
	• Accumulate Total-Hotel-Revenue.	

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IPO chart (Input - Processing - Output)

Input	Processing	Output
• Season	• Accept guest reservation inputs.	• Individual Guest Bill Amount (Final-Price)
• Room-Type	• Determine Room Rate	• Cumulative Hotel Revenue
• Nights	• Multiply Rate by Nights.	(Total-Hotel-Revenue)
	• Apply 15% discount if Nights > 7.	
	• Compute Final-Price	
	• Add Final-Price to running revenue total.	

Pseudocode:

BEGIN

DECLARE Total-Hotel-Revenue, N, Counter, Season, Room-Type, Nights, Rate, Initial Price, Discount, Final Price.

FOR Counter = 1 to N DO

PRINT "Enter Details for Guest", Counter.

INPUT Season, Room-Type, Nights.

IF Season == "Peak" THEN

IF Room-Type == "Standard" THEN Rate = 5000

ELSEIF Room-Type == "Deluxe" THEN Rate = 8000

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ELSEIF Room-Type == "Suite" THEN Rate = 12000
ENDIF

ELSE

IF Room-Type == "Standard" THEN Rate = 3000

ELSEIF Room-Type == "Deluxe" THEN Rate = 5000

ELSEIF Room-Type == "Suite" THEN Rate = 8000.

ENDIF

Initial-Price = Rate * Nights

IF Nights > 7 THEN

Discount = Initial-Price * 0.15

ELSE

Discount = 0

ENDIF

Final-Price = Initial-Price - Discount

Total-Hotel-Revenue = Total-Hotel-Revenue +
Final-Price

PRINT "Final price for guest: Rs. ", Final-
Price

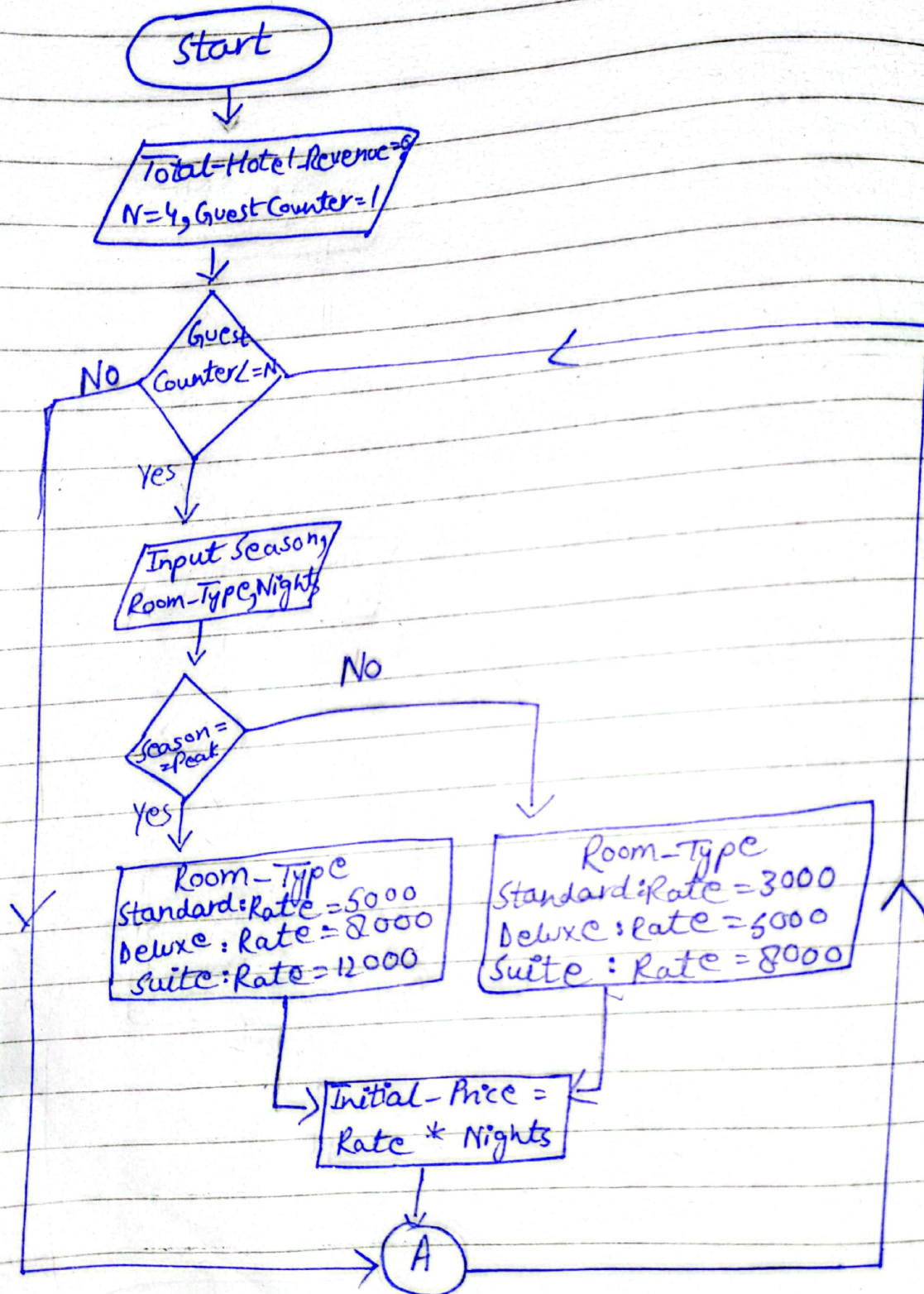
ENDFOR

PRINT "Total cumulative Hotel Revenue: Rs. ",
Total-Hotel-Revenue

END

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Flowchart:



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